Magnetostratigraphy of Quaternary and late Tertiary sediments on Banks Island, Canadian Arctic Archipelago[1]

R.W. Barendregt, J.-S. Vincent, E. Irving, and J. Baker

Abstract: Sediments approximately 50 m thick from Banks Island (Canadian Arctic Archipelago) contain one of the longest terrestrial records of Pleistocene climate changes in North America. Samples have been obtained from 126 horizons distributed among four localitites, of which 116 horizons yielded acceptable paleomagnetic data. In sediments of the Matuyama Reversed Zone, there are recorded at least two and possibly as many as five full continental glaciations, two interglacial intervals, and a nonglacial interval at the beginning which is considered preglacial. Subzones attributable to the Olduvai and Jaramillo are present within the Matuyama Reversed Zone. The Brunhes Normal Zone records three full continental glaciations and three interglaciations. The Brunhes–Matuyama boundary occurs within interglacial deposits. The preglacial Worth Point Formation records a climate milder than today, and cooler than that of the late Tertiary. Based on floral, faunal, stratigraphic, and paleomagnetic constraints, a normal polarity sequence in the Worth Point Formation is assigned to the Olduvai normal polarity subzone (1.95–1.77 Ma). The earliest direct evidence of glaciation on Banks Island occurs in sediments that postdate the Worth Point Formation (<1.77 Ma). Consequently, in the western Canadian Arctic, the first continental glaciation postdated the first glaciation in the Canadian Cordillera (2.6 Ma) by at least a million years. The overall mean direction of the Quaternary geomagnetic field in Banks Island does not differ significantly from the geocentric axial dipole field, and these sediments contain no inclination error.

Résumé : Les sédiments d'une puissance approximative de 50 m sur l'île Banks (archipel arctique canadien), constituent un des plus longs registres terrestres de changements climatiques survenus durant le Pléistocène en Amérique du Nord. Les échantillons ont été prélevés dans 126 horizons, répartis en quatre localités; 116 de ces horizons ont fourni des données paléomagnétiques fiables. Dans les sédiments de la Zone inversée de l'époque Matuyama, on y trouve d'enregistrer au moins deux, et peut-être jusqu'à cinq, glaciations continentales complètes, deux intervalles interglaciaires et un intervalle non -glaciaire marquant le début qui est interprété comme préglaciaire. En plus, on reconnaît dans la Zone inversée de l'époque Matuyama des sous-zones qui peuvent être attribuées aux épisodes Olduvai et Jaramillo. La Zone normale de l'époque Brunhes a enregistré trois glaciations continentales complètes et trois intervalles interglaciaires. La limite Brunhes–Matuyama apparaît dans les dépôts interglaciaires. La Formation de Worth Point préglaciaire révèle l'existence d'un climat plus tempéré que celui d'aujourd'hui, et plus froid que celui du Tertiaire tardif. Les études de la flore, la faune, la stratigraphie et les contraintes paléomagnétiques suggèrent qu'une séquence de polarité normale dans la Formation de Worth Point puisse être assignée à la sous-zone de polarité normale d'Olduvai (1,95–1,77 Ma). Les sédiments qui postdatent la Formation de Worth Point (<1,77 Ma) fournissent une évidence directe de la plus ancienne avancée glaciaire sur l'île Banks. Donc, dans l'ouest de l'Arctique canadien, la première glaciation continentale est post-datée d'au moins un million d'années relativement à la première glaciation survenue dans la Cordillère au Canada (2,6 Ma). La direction moyenne globale du champ géomagnétique quaternaire dans l'île Banks, ne diffère pas significativement du champ du dipole axial géocentrique, et ces sédiments ne comportent pas d'erreur d'inclinaison.
[Traduit par la rédaction]

Introduction

Stratigraphic sections on Banks Island, in the southwestern Canadian Arctic Archipelago (Fig. 1), provide a record of multiple continental glaciations and interglaciations. All the ice sheets originated from distant sources located to the southeast of Banks Island on the Canadian mainland. The purpose of this study was to apply paleomagnetism as a tool to define better the chronology of events, and to relate them to the global polarity time scale and the oceanic isotope record.

Vincent (1983, 1984, 1989, 1990, 1992) reconstructed a sequence of nonglacial and glacial events (Table 1). At the base, he identified a nonglacial interval recorded by the Worth Point Formation. No glaciogenic sediments occur within or below the Worth Point Formation, which he therefore considered to have been deposited prior to the onset of glaciation. Glaciation began with the extensive continental Banks Glaciation. Then followed a nonglacial interval (Morgan Bluffs Interglaciation), another glaciation (Thomsen Glaciation), an interglacial interval (Cape Collinson Interglaciation), and

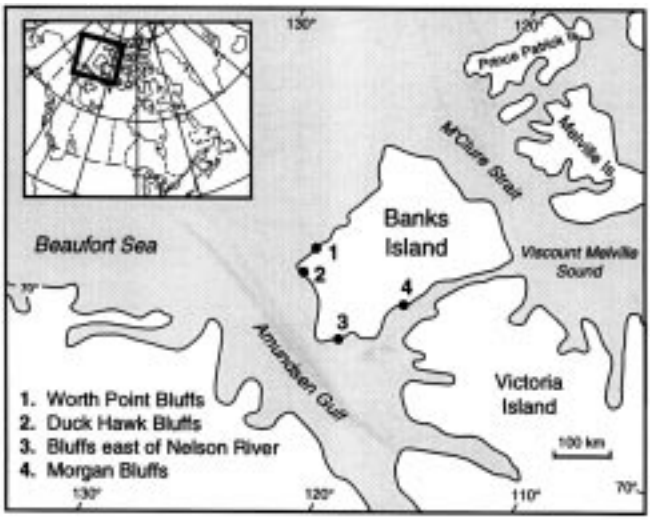
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Fig. 1. Location map showing localities studied on Banks Island.



finally the eastern part of Banks Island was covered by ice one last time (Amundsen Glaciation). Interglacial conditions prevail at present.

The record of these events is best preserved at four localities: Worth Point, Duck Hawk, Nelson River, and Morgan bluffs (Fig. 1). Paleomagnetic results described previously from Duck Hawk Bluffs allow the assignment of preglacial Worth Point Formation, sediments associated with the Banks Glaciation, and interglacial beds of the overlying Morgan Bluffs Formation to the Matuyama Reversed Zone (Vincent et al. 1984; Barendregt and Vincent 1990). The uppermost beds in the Morgan Bluffs Formation revealed a reversal transition to the Brunhes Normal Zone. The overlying deposits of the Thomsen Glaciation, Cape Collinson Interglaciation, and Amundsen Glaciation were normally magnetized and therefore were assigned to the Brunhes Normal Zone. In this paper, we report new paleomagnetic data from the other three localities and combine them with earlier data from Duck Hawk Bluffs to provide a general magnetostratigraphic framework for the sequence of glacial and nonglacial events on Banks Island. This framework is given in Table 2, in which, anticipating the results described below, the sequence of glacial and nonglacial events recorded at the four localities is listed.

Paleomagnetic procedures

Although marine wave-cut bluffs provide relatively easy access, there are many sampling limitations. Sections are affected by permafrost, are sometimes disturbed by glaciotectonic processes, and in places are scree covered. Careful selection and cleaning of outcrops is therefore required. The common occurrence of very coarse grained deposits, particularly in tills, limits any attempt to obtain continuous vertical sampling. At Worth Point Bluffs and Nelson River Bluffs, all units occur in a single vertical exposure. At Duck Hawk Bluffs and Morgan Bluffs, several sections were sampled from which composite records were compiled on the basis of sedimentological, stratigraphic, and magnetic polarity. Notwithstanding these limitations, it has been possible to establish a fairly complete and consistent magnetostratigraphy for the four localities.

For the present work, samples were collected in 1983, 1988, and 1992. Sections were, in many instances, cleaned down to permafrost with jets of water pumped from the sea, left to thaw for a time, and then trimmed to a vertical face by shovel and trowel. Plastic holders (cubes in early collections, and cylinders in later collections) were then pressed into the just unfrozen sediment and oriented by Brunton compass checked daily against solar bearings. New data are described from one section at Worth Point Bluffs (Fig. 2), one at Nelson River Bluffs (Fig. 4), and six at Morgan Bluffs (Fig. 5). Previous data from seven sections at Duck Hawk Bluffs (Fig. 3) are recompiled herein.

Natural remanent magnetization (NRM) was measured on automated Schonstedt spinner magnetometers. Twelve readings were obtained from each specimen, yielding four determinations of direction from which within-specimen precision k_{sn} was calculated (Irving et al. 1985); k_{sn} is a measure of inhomogeneity of natural remanent magnetization within specimens (k_s is a measure of inhomogeneity of specimens after demagnetization). Pilot specimens were selected from each section and demagnetized at 10 steps up to 100 mT. Based on results from these, all other specimens were then demagnetized at selected levels to determine the components present (Figs. 6–10).

NRM intensities varied through three orders of magnitude. Intensities at Worth Point and Duck Hawk bluffs ranged from 10^{-4} to 10^{-3} A/m, and at Nelson River and Morgan bluffs from 10^{-3} to 10^{-2} A/m. This increase from east to west is compatible with a source of detritus from Proterozoic mafic dykes and lavas of Victoria Island and other up-ice igneous rocks to the southeast. Many specimens have intensities approaching the limit of detection of the magnetometers (10^{-4} A/m). Values of k_{sn} were highly variable, ranging from 2 to 150. Specimens within which NRM was essentially randomly distributed ($k_{sn} < 5$) were rejected. During demagnetization, k_s values generally decrease, often dramatically, as intensity falls, and when k_s fell below 5 processing was stopped and the specimen rejected. Altogether 134 specimens, or 15% of the 908 specimens collected, were rejected on the grounds that they contain no discernible record of the geomagnetic field. Of the samples that yielded acceptable data, 45% were taken from beds of silts and clays (including loess and rhythmites), 34% from tills, and 18% from fine sands.

The major objective was to establish a polarity stratigraphy. A second objective was to obtain mean directions and paleopoles for Quaternary time, for which more precise directions of magnetization are required. The identification of polarity was made in this way. If, during demagnetization, the inclinations remained positive and ≥45°, a specimen was designated as having normal polarity (N); if, during demagnetization, inclination remained or became negative and ≥–45°, a specimen was designated reversed (R). N and R magnetizations that have comparatively high internal precisions ($k_s \geq 29$, corresponding to a standard deviation <15°) are designated N*, R*. N* and R* specimens, which are always silts or clayey silts, typically have excellent end points and decayed linearly to the origin on orthogonal diagrams. They are considered to be good recorders of the geomagnetic field which can be used to calculate mean directions and paleopoles. N and N*, R and

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Table 1. Previous correlation of lithostratigraphic units and glacial and interglacial events on Banks Island (modified from Vincent 1992).

General Chronostratigraphy			Geological Events		Stratigraphy
HOLOCENE			Interglaciation		
10 ka					
LATE PLEISTOCENE	Late	WISCONSINAN	(Russell Stade)	Prince of Wales Fm.	Schuyter Point Sea seds. Passage Point seds. Schuyter Point Sea seds.
23 ka					
	Middle		Amundsen Glaciation		Nonglacial seds.?
65 ka					
	Early		(M'Clure Stade)		Carpenter Till East Coast, Meek Point, Investigator Sea seds. Jesse, Sachs, Bar Harbour, Mercy tills
80 ka					Pre-Amundsen Sea seds.
SANGAMONIAN			Cape Collinson Interglaciation		Cape Collinson Formation
130 ka					
MIDDLE PLEISTOCENE			Thomsen Glaciation	Nelson River Fm.	Big Sea seds. Lake Parker and Lake Dissection seds. Kellett Till, Baker Till, and Kange Till Pre-Thomsen Sea seds.
Brunhes / Matuyama	780 ka		Morgan Bluffs Interglaciation		Morgan Bluffs Formation
EARLY PLEISTOCENE			Banks Glaciation	Duck Hawk Bluffs Fm.	Post-Banks Sea sediments Lake Engina and Lake Storkerson sediments Bernard Till, Plateau Till, and Durham Heights Till Pre-Banks Sea sediments
			Erosion and sedimentation during preglacial interval		Worth Point Formation
1.61 Ma			Early glaciation(s) ?		
Matuyama / Gauss	2.58 Ma		? ?		?
LATE TERTIARY			Fluvial sedimentation on coastal plain		Beaufort Formation

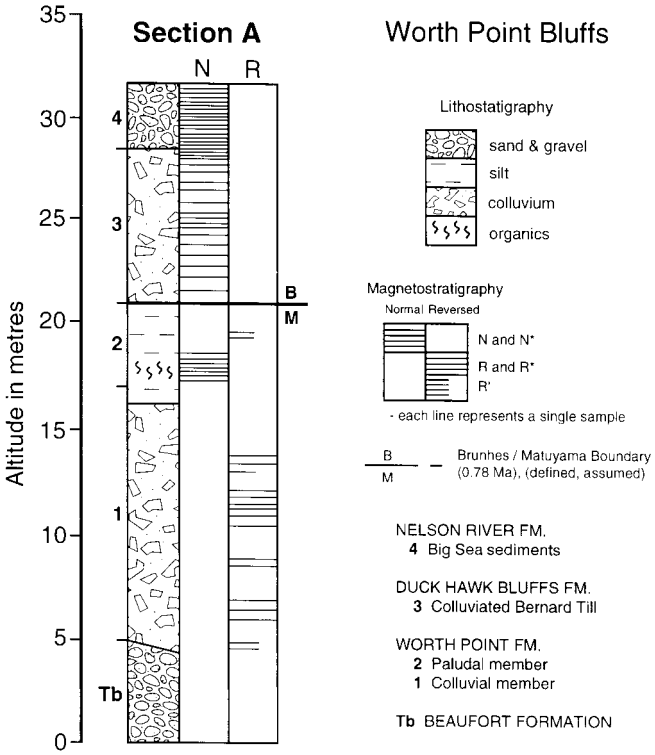
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Table 2. Correlation of Pleistocene glacial and interglacial events at four localities on Banks Island.

			WORTH POINT BLUFFS	DUCK HAWK BLUFFS	NELSON RIVER BLUFFS	MORGAN BLUFFS	MAJOR ★ POLARITY UNITS
QUATERNARY	LATE PLEISTOCENE	BRUNHES	Present Interglaciation *	Present Interglaciation (11)	Present Interglaciation *	Present Interglaciation *	7
				Amundsen Glaciation (9,10)	Amundsen Glaciation (12,13)	Amundsen Glaciation (13)	
				Cape Collinson Interglaciation (8)	Cape Collinson Interglaciation (11)		
		0.13					
	MIDDLE PLEISTOCENE		Thomsen Glaciation (4)	Thomsen Glaciation (7)	Thomsen Glaciation (9,10)	Thomsen Glaciation (11,12)	6
				Unnamed Interglaciation (6)	Unnamed (8) Interglaciation		5
					Unnamed (6,7) Glaciation		4
		0.78		Unnamed Interglaciation (5)		Unnamed Glaciation (7,8,9,10)	
		0.99					
		JARAMILLO 1.07				Morgan Bluffs Interglaciation (6)	3
	EARLY PLEISTOCENE	MATUYAMA	Banks Glaciation *	Banks Glaciation (2,3,4)	Up to 4 Glaciations including Banks Glaciation (1,2,3,4,5)	Up to 2 Glaciations including Banks Glaciation (1,2,3,4,5)	2
		1.61					
		1.77	Preglacial WORTH POINT FORMATION (1,2)	Preglacial WORTH POINT FORMATION (1)			1
		OLDUVAI 1.95					
TERTIARY							

(3) numbers in parentheses refer to unit numbers in Figs. 2,3,4,5, and 12. Numbers in **bold** are till sheets
* event is recorded in vicinity of bluffs
★ defined in text

Fig. 2. Lithostratigraphy and magnetostratigraphy of the Worth Point Bluffs.

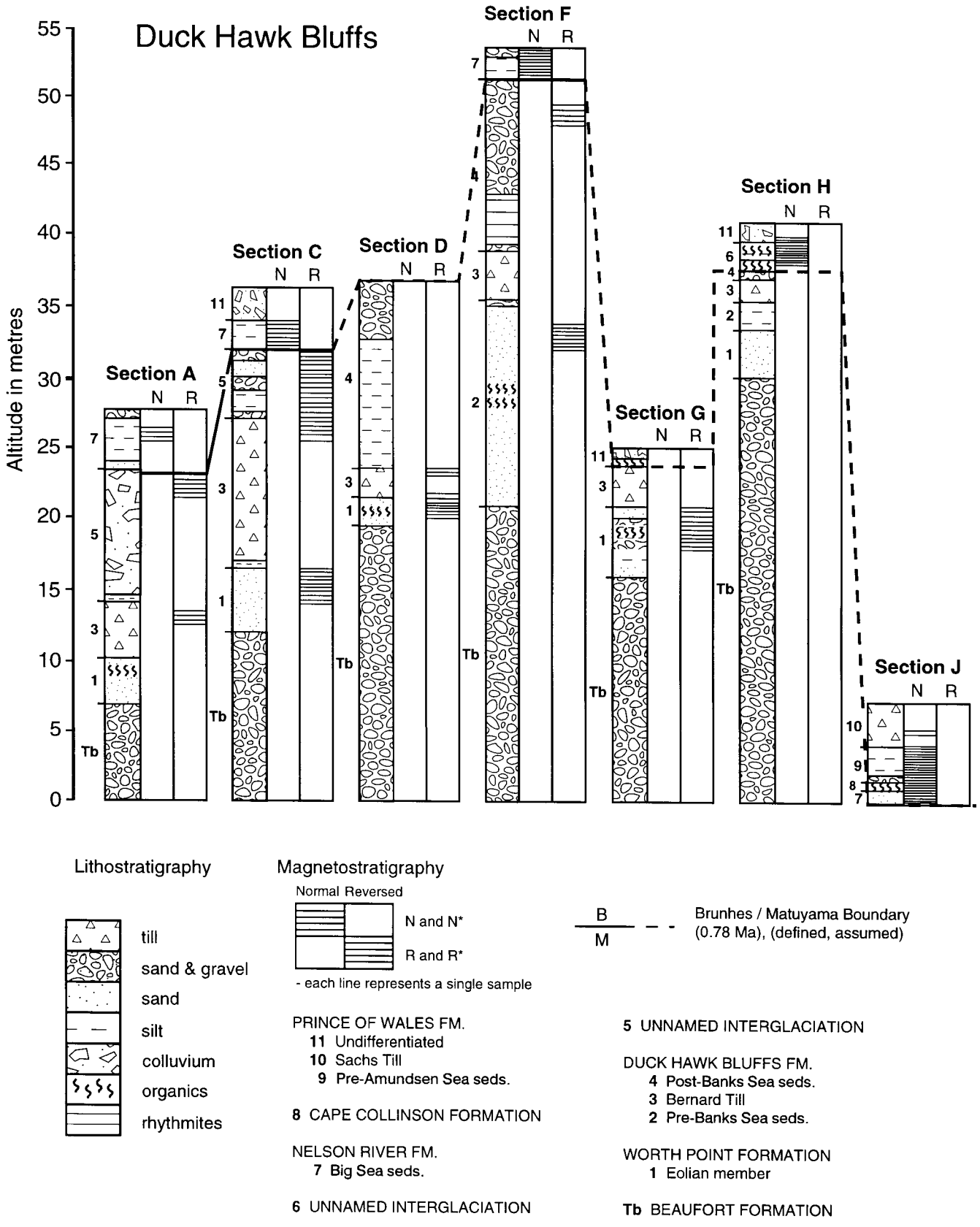


R* magnetizations provide good indicators of polarity. In addition, 93 specimens, mainly from the lower parts of sections, migrated systematically away from the present Earth's field during demagnetization but did not reach end points. We interpret such behaviour as indicative of incomplete removal of Brunhes-age overprints and of the presence of an underlying reversed polarity; such specimens are designated R′ and are typically coarse grained, largely from tills. A determination of R′ from a single specimen is not in itself regarded as a reliable indication of reversed polarity, but if several are observed in sequence then the assignment of reversed polarity is made with some confidence. The distribution of the different types of magnetization is given in Table 3.

Examples are shown in Figs. 6–10. Figure 6 shows the demagnetization of an N* specimen from interglacial marine deltaic silt (unit 8) at Nelson River Bluffs. The magnetization is strong and its decay smooth. The directions remain approximately constant throughout, that is, there is a good end point. Figure 7 shows the demagnetization of an R* specimen from marine rythmites (unit 1, section H) at Morgan Bluffs. The magnetization is again strong, and there is a good end point and a well-defined reversed polarity. Figure 8 shows an R magnetization in a preglacial colluvial unit (unit 1) from the lower part of the Worth Point Formation. The intensity of magnetization is lower and the directions are more scattered, but they clearly indicate reversed polarity. Figure 9 shows R′ magnetization in interglacial fluvial sands from unit 6 (section L)

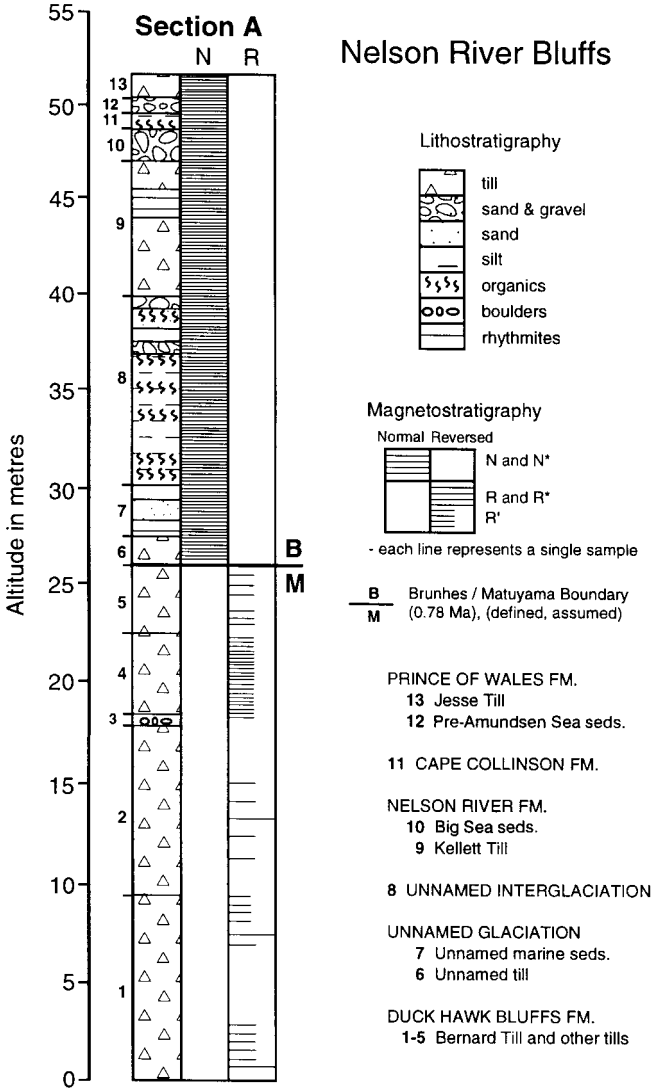
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Fig. 3. Lithostratigraphy and magnetostratigraphy of the Duck Hawk Bluffs.



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Fig. 4. Lithostratigraphy and magnetostratigraphy of the Nelson River Bluffs.



at Morgan Bluffs. The NRM has normal polarity, but during demagnetization the inclinations become negative without reaching the reversed end points typical of R* magnetization; as already noted, such a result in isolation does not necessarily imply reversed polarity, but if several are observed in sequence, reversed polarity can be inferred. A further example of R′ magnetization from a till (unit 2) at Nelson River Bluffs is shown in Fig. 10. The R and R′ magnetizations of Figs. 8–10 consist of a strong Brunhes overprint with a weak underlying reversed magnetization.

Lithostratigraphy and magnetostratigraphy of measured sections

Worth Point Bluffs

Vincent (1990) identified four units resting on late Tertiary fluvial coastal plain sediments of the Beaufort Formation (Fig. 2, Tb). At the base are preglacial colluvial (unit 1) and paludal (unit 2) sediments of the Worth Point Formation preserved as local valley fill. They contain rich in situ paleofauna and flora, and conditions were sufficiently warm to allow growth of conifers. These preglacial deposits are in turn overlain by a diamicton (unit 3), which Vincent considered to be derived from the Bernard Till of the Banks Glaciation on the basis of their similar pebble and matrix lithology. This diamicton has been extensively reworked by periglacial processes (see below). The sequence is capped by sands and gravels (unit 4) laid down in the Big Sea, a marine transgression which covered extensive areas of Banks Island at the time of the Thomsen Glaciation.

Samples from 11 m of colluvial sediments (unit 1) are reversed and assigned to the Matuyama Chron (Fig. 2). The overlying 5 m thick upper paludal member of the Worth Point Formation (unit 2) exhibits both polarities. At the base, thick peat mats with silts are normally magnetized, whereas overlying interstratified thinner peat mats and silts are reversed.

The diamicton of unit 3 is normal (Fig. 2). Laterally extensive and overlying Big Sea sands and gravels (unit 4), associated with the Thomsen Glaciation, are also normal. The Bernard Till at the nearby Duck Hawk Bluffs locality is reversed and of Matuyama age (see below; Barendregt and Vincent 1990). The diamicton at Worth Point Bluffs is normal (unit 3, Fig. 2), but, as observed in a new exposure in 1992, it has been severely disrupted by later ground-ice growth. When this happens, the disrupted body is in part reduced to a slurry, the original magnetization is destroyed, and a new magnetization is acquired as the magnetic grains rotate into the geomagnetic field (Irving and Major 1964). Hence, the magnetization of the Worth Point diamicton is probably an overprint resulting from extensive reworking by periglacial processes.

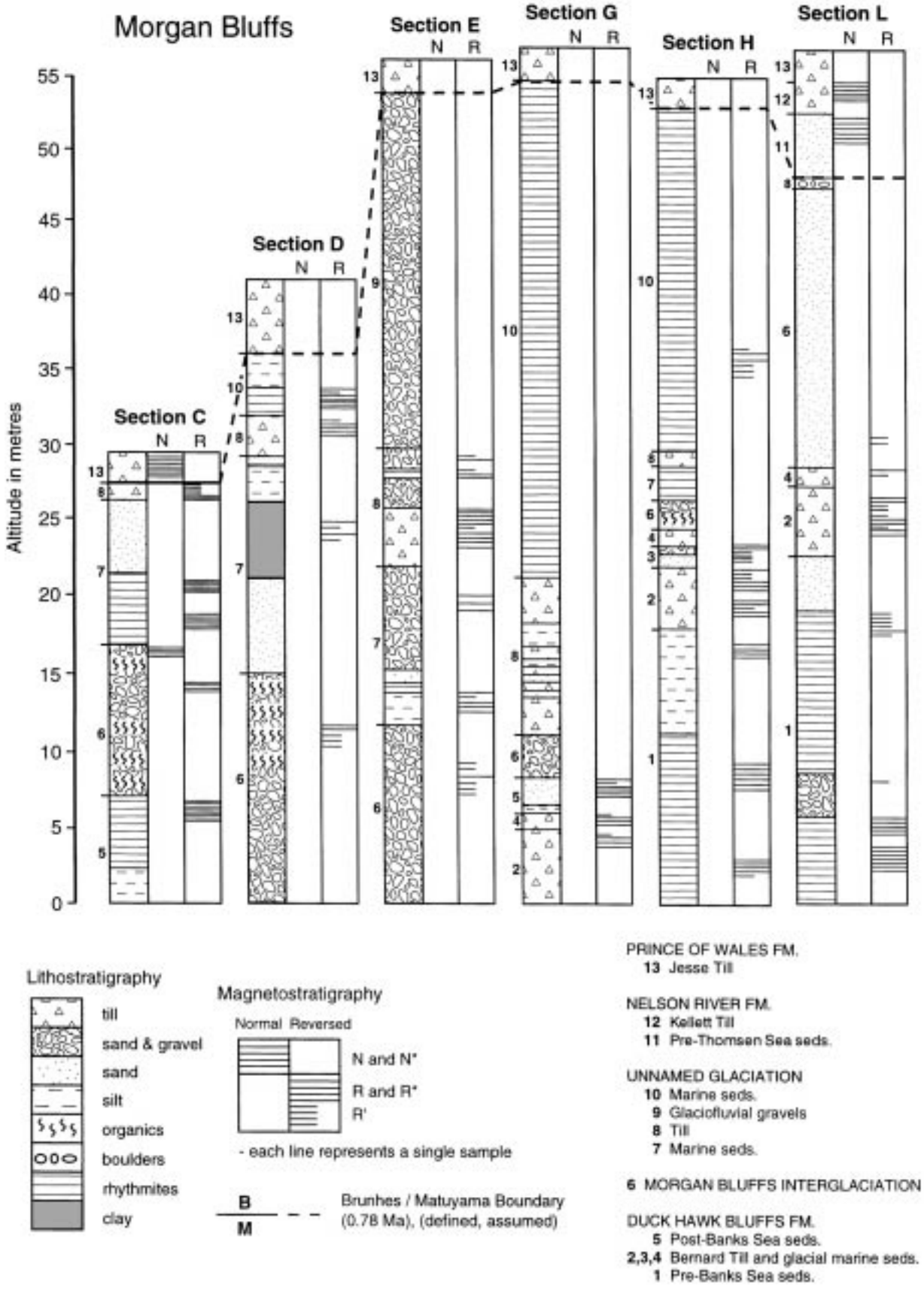
The colluvial and paludal members of the Worth Point Formation lie entirely within the Matuyama Chron. The overlying diamicton and marine sands and gravels (units 3 and 4) are assigned to the Brunhes Chron. The organic sediments with normal polarities within the paludal member could belong to either the Jaramillo or Olduvai subzones. They are assigned to the Olduvai subchron (1.95–1.77 Ma; Cande and Kent 1995) for two reasons. They were laid down in a warm climate, typical of the latest Pliocene or earliest Pleistocene (Vincent 1990), rather than the cooler conditions of the later early Pleistocene in which the Jaramillo (1.07–0.99 Ma) is located. Second, as discussed later, assignment to the Jaramillo would only allow 80 ka during the Matuyama Chron to accommodate at least two and possibly as many as five full glaciations, and at least two interglaciations, all of which are recorded elsewhere on Banks Island after the deposition of the Worth Point Formation. Eighty thousand years is, we believe, insufficient.

Duck Hawk Bluffs

This locality (Figs. 1, 3) contains 11 lithostratigraphic units (Vincent et al. 1983; Matthews et al. 1986; Vincent 1990). Aeolian sediments with associated pond deposits (unit 1), assigned to the Worth Point Formation, lie on fluvial late Tertiary Beaufort coastal plain sands and gravels of the Beaufort Formation. These are overlain by pre-Banks Sea sediments (unit 2), Bernard Till (unit 3), and post-Banks Sea sediments (unit 4), all associated with the Banks Glaciation. Above these are terrestrial nonglacial sediments, initially assigned to the Morgan Bluffs Interglaciation (Table 1) (Vincent 1992).

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Fig. 5. Lithostratigraphy and magnetostratigraphy of the Morgan Bluffs.



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Fig. 6. N* magnetization in interglacial perimarine silts (unit 8) at the Nelson River Bluffs locality. Top shows the decay of intensity with increasing alternating field; bottom shows corresponding directions, plotted relative to present horizontal. ●, positive inclinations.

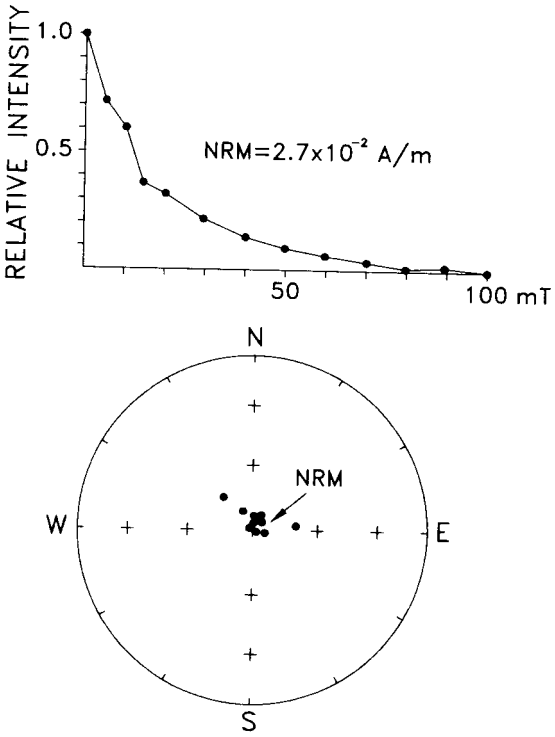


Fig. 7. R* magnetization in marine rhythmites (unit 1, section H) of the Morgan Bluffs locality. Top shows the decay of intensity with increasing alternating field; bottom shows corresponding directions, plotted relative to present horizontal. ● and ○, positive and negative inclinations, respectively.

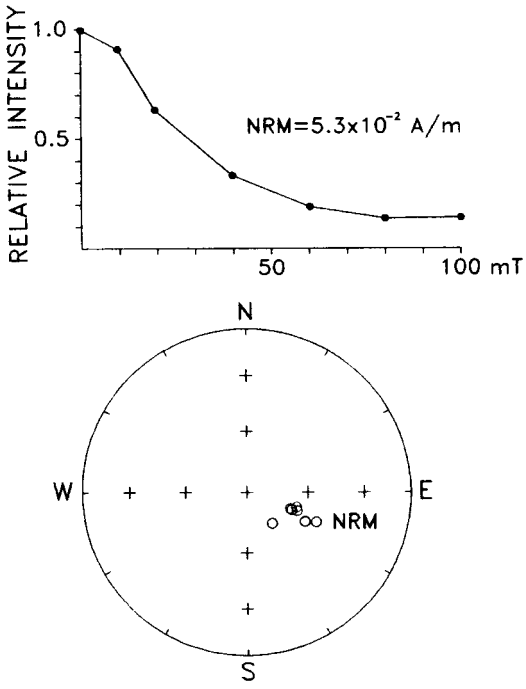
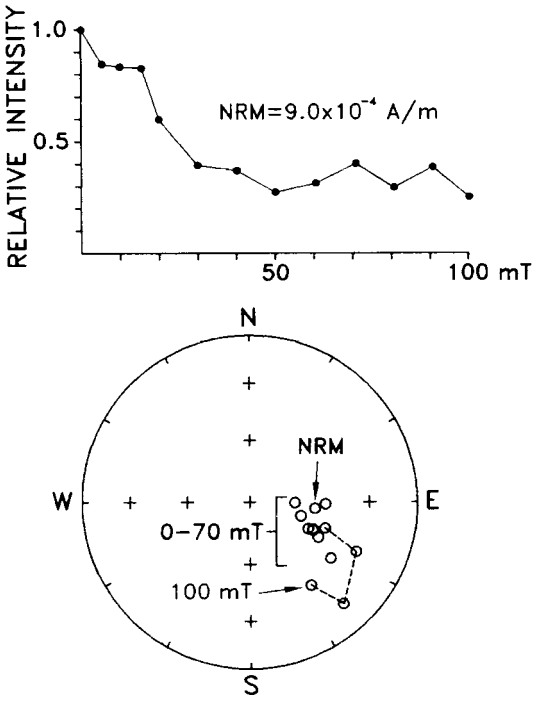


Fig. 8. R magnetization in colluvial member (unit 1) of the Worth Point Formation at the Worth Point Bluffs locality. See Fig. 7 caption.



These are now subdivided into two distinct interglacial units: (*i*) tundra pond, colluvial, fluvial, aeolian, and possibly marine deposits recording terrestrial periglacial interglacial conditions (unit 5, sections A and C); and (*ii*) wind-blown silts associated with peat beds (unit 6, section H) recording a younger interglaciation. Marine sediments (unit 7) of the Big Sea (associated with the Thomsen Glaciation), woody peats (unit 8) assigned to the Cape Collinson Interglaciation, glaciomarine sediments of the pre-Amundsen Sea (unit 9), Sachs Till of the Thesiger Lobe of the Amundsen Glaciation (unit 10), and thin undifferentiated Holocene deposits (unit 11) complete the sequence.

Ten units (Vincent et al. 1984; Barendregt and Vincent 1990) were sampled in seven sections (Fig. 3). Unit 11, a stony till, could not be sampled. Units 1–4 and the lower interglacial strata of unit 5 are reversed, and they are assigned to the Matuyama Chron. Unit 5 records a polarity transition from reverse to normal. The overlying interglacial units 6 and 7 (sections A, C, F, and H, Fig. 3) are normal and are assigned to the Brunhes Chron. The Brunhes–Matuyama boundary occurs in the upper beds of the interglacial sequence of unit 5 (section C) which overlie deposits laid down during the Banks Glaciation (units 2–4). Sedimentation within unit 5 appears to be uninterrupted, and a relatively rare instance of the Brunhes–Matuyama transition in a terrestrial sequence appears to be recorded. Because of their differing polarities, these interglacial deposits of unit 5, which straddle the Brunhes–Matuyama boundary, cannot, as previously thought (see Table 1), be correlated with the deposits assigned to the Morgan Bluffs Interglaciation at the Morgan Bluffs type section. There, as will be discussed later, a reversely magnetized sequence of deposits

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Fig. 9. R′ magnetization in marine sands (unit 6, section L) at Morgan Bluffs locality. See Fig. 7 caption.

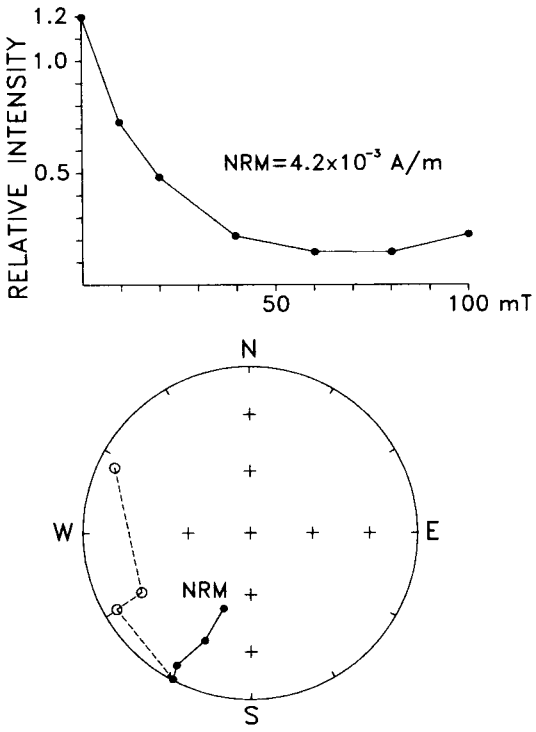


Fig. 10. R′ magnetization in till from unit 2 of the Nelson River Bluffs locality. See Fig. 7 caption.

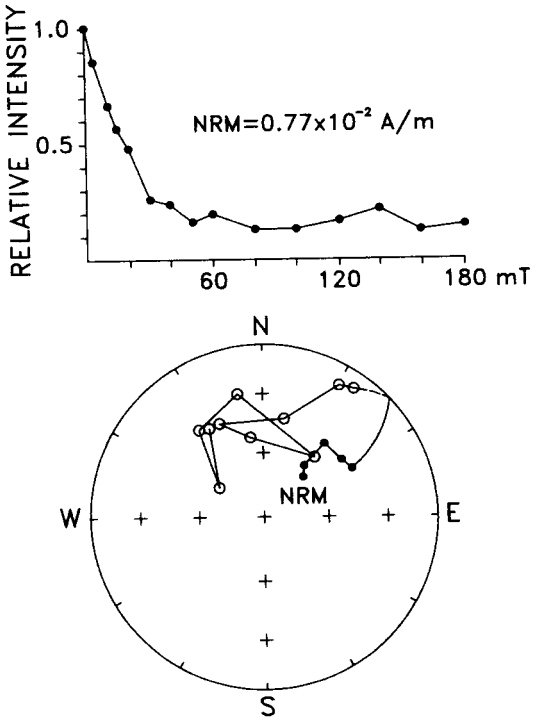


Table 3. Distribution of types of magnetization.

Quaternary sediments	$k < 5$	R′	R	N	R*	N*
Nelson River Bluffs	39	25	11	175	—	14
Worth Point Bluffs	41	12	20	43	7	33
Morgan Bluffs	54	56	119	30	78	25
Total	134	93	150	248	85	198

Notes: Specimens in which the precision of NRM (k_{sn}) is less than 5 or falls below 5 during demagnetization (k_s) have been rejected and are listed in the first column. The other five categories are explained in the text.

occurs which records a continental glaciation that postdates the Morgan Bluffs Interglaciation.

Nelson River Bluffs

At this locality near the southernmost point of Banks Island (Fig. 1), Vincent (1983, 1990) has described 13 lithostratigraphic units (Fig. 4). At the base, there are five tills (units 1, 2, 4, 5, and 6) with variable composition, fabric, and colour. At least one, but possibly as many as five distinct continental glaciations are represented, all of which have been assigned previously to the Banks Glaciation. Contacts between individual till sheets are well marked. For example, the tills of units 2 and 4 are separated by a continuous boulder pavement (unit 3) which suggests, but does not prove, that multiple glaciations occurred. Overlying these till sheets are marine sediments (unit 7) likely associated with a marine transgression (post-Banks Sea) which followed the glaciation responsible for depositing the till of unit 6 (Table 1).

The upper half of the section begins with a 9 m thick interglacial sequence of interstratified fine sediments and in situ peat layers, representing deposition in a perimarine delta (unit 8), and the Kellett Till and the Big Sea rhythmites and deltaic sands and gravels (units 9 and 10), both deposited during the Thomsen Glaciation (Nelson River Formation). In situ peats and shrubs developed on a former tundra surface during the succeeding Cape Collinson Interglaciation (unit 11). Finally, pre-Amundsen Sea sediments and Jesse Till (units 12 and 13), both associated with the Amundsen Glaciation and referred to the Prince of Wales Formation, were deposited.

Oriented samples were collected from all units, except the boulder pavement (unit 3) separating the till sheets of units 2 and 4. The distinct till units 1, 2, 4, and 5 have NRMs that are well grouped, with steep positive inclinations and easterly declinations close to the present Earth's field. Demagnetization generally yields scattered but reversed inclinations (Fig. 10; R′ half bars in Fig. 4). Three samples from units 1 and 2 have clearly defined reversed directions (R full bars in Fig. 4). All samples have large Brunhes-age overprints, with small underlying reversed magnetizations. On this basis, the lower part of the Nelson River section is assigned to the Matuyama Reversed Chron.

Glacial and nonglacial sediments from units 6–13, from the upper half of the section, have NRM directions tightly clustered almost vertically downward. Magnetizations remain normal throughout the demagnetization range, and directions remained little changed up to and often beyond 50 mT (Fig. 6). No reversed magnetizations were found. The upper part of the section is therefore assigned to the Brunhes Normal Chron. The Brunhes–Matuyama boundary lies between the till sheets of units 5 and 6; therefore, these tills belong to different polarity zones, implying that they are separate lithostratigraphic units and represent separate glaciations. The results also indi-

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Fig. 11. Ice-wedge cast developed in Morgan Bluffs interglacial gravels (Nelson River Formation), unit 6, section C, Morgan Bluffs locality. Sediments in ice-wedge cast have normal polarity, and are assigned to Jaramillo polarity subchron (1.07–0.99 Ma). Sediments above and below ice-wedge cast are reversely magnetized.



cate two Middle Pleistocene glaciations represented by units 6 and 7 and 9 and 10. The normally magnetized interglacial deposits of unit 8 cannot be associated with sediments assigned to the Morgan Bluffs Interglaciation (as previously thought, Table 1), as these are reversely magnetized at the type section (described in the following section).

Morgan Bluffs

At Morgan Bluffs Vincent (1983, 1990) recognized 13 lithostratigraphic units (Figs. 1, 5) distributed among many sections, of which sections C, D, E, G, H, and L were sampled. As a result of fieldwork in 1992, minor stratigraphic revisions have been made and are incorporated herein.

Several units record the first continental ice advance onto Banks Island and are assigned to the Banks Glaciation: the pre-Banks Sea sediments (unit 1), Bernard Till and glaciomarine sediments (units 2–4), and post-Banks Sea sediments (unit 5) collectively are assigned to the Duck Hawk Bluffs Formation. These beds are overlain sequentially by interglacial fluvial sediments with in situ peat layers (unit 6, Morgan Bluffs Formation), a suite of marine sediments (unit 7), a till (unit 8), glaciofluvial gravels (unit 9), and a marine regressive sequence (unit 10), all associated with an unnamed continental glaciation; pre-Thomsen Sea sediments (unit 11) and Kellett Till (unit 12) of the Thomsen Glaciation (Nelson River Formation); and finally by the Jesse Till (unit 13) of the Amundsen Glaciation (Prince of Wales Formation).

Oriented samples were collected from six sections (C, D, E, G, H, L). Suitable sediments occurred in all but one of the units. Sediments associated with the Banks Glaciation (units 1–5), the Morgan Bluffs Interglaciation (unit 6), and an unnamed glaciation (units 7–10 originally assigned to the Middle Pleistocene Thomsen Glaciation) all exhibited reversed polarity, and are assigned to the Matuyama Chron. The uppermost units (11 and 12) associated with the Thomsen Glaciation and unit 13 associated with the Amundsen Glaciation are normal and are assigned to the Brunhes Chron. The Brunhes–Matuyama boundary, which lies within the Morgan Bluffs locality, postdates the unnamed glaciation (units 7–10) and predates the Thomsen Glaciation (units 11 and 12).

Fine-grained deposits, which fill fossil ice-wedge casts (Fig. 11) penetrating the uppermost part of the interglacial sediments of the reversely magnetized Morgan Bluffs Formation (unit 6, section C), have normal polarity. These fossil ice wedges were formed by periglacial processes that affected the tundra surface at that time. These wedges were buried by reversely magnetized sediments of unit 7. Hence, they record an interval of normal polarity within the Matuyama Chron. They are assigned to the Jaramillo (1.07–0.99 Ma) rather than to the Olduvai subzone, because the Morgan Bluffs Interglacial sequence, at Morgan Bluffs, is both underlain and overlain by deposits recording full continental glaciations (units 1–5 and 7–10), indicating that the deposits postdate the warmer Late Pliocene – Early Pleistocene interval documented by the Worth Point Formation (Table 2; Figs. 12, 13). The normal subzone recorded within "warm" preglacial Worth Point sediments (assigned to the Olduvai) is clearly different from the normal subzone recorded from "cold" ice-wedge casts within the Morgan Bluffs Formation (assigned to the Jaramillo).

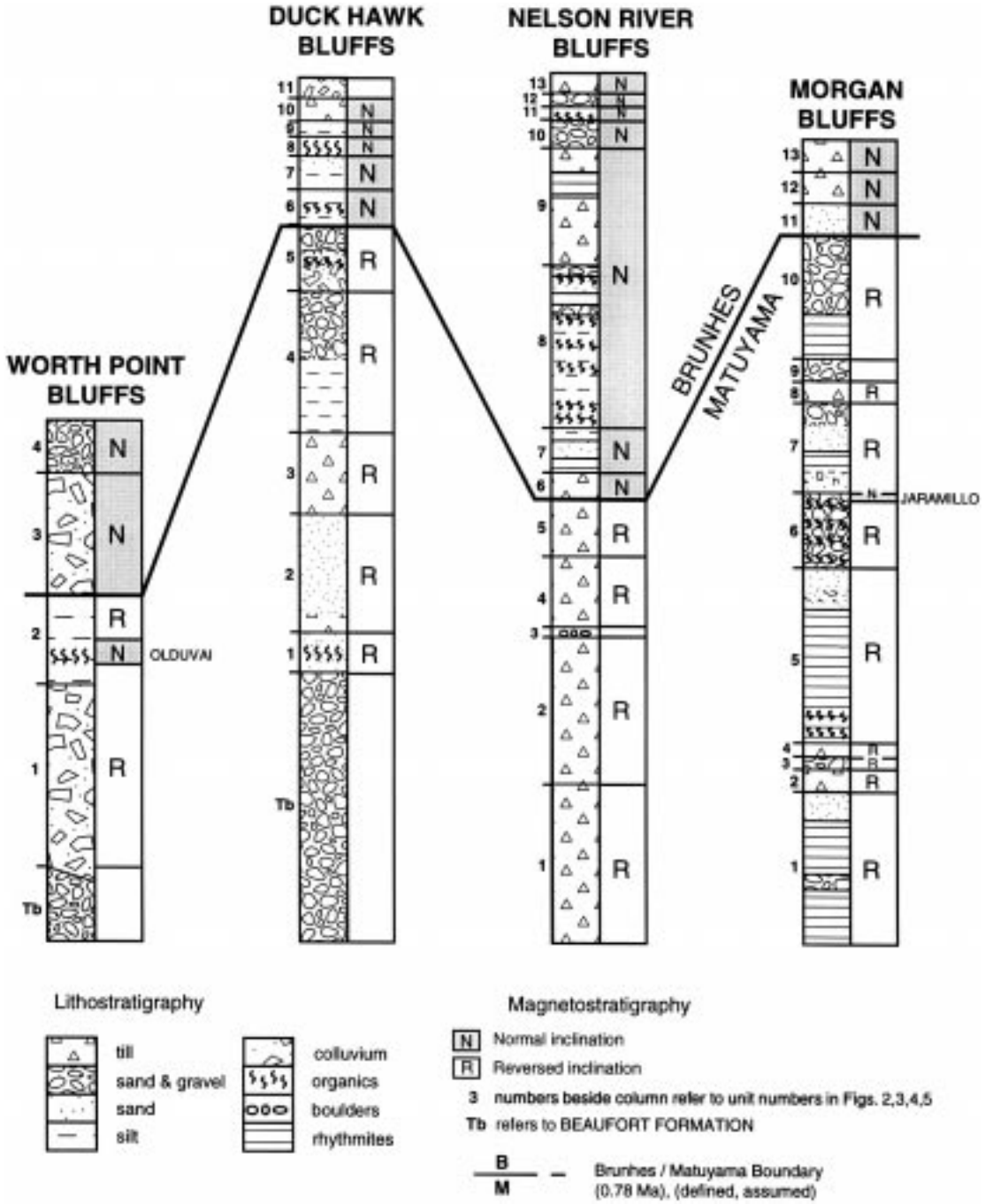
Correlation of composite sections and chronostratigraphy

Composite sections showing the lithostratigraphic units and their polarities are given in Fig. 12. The Brunhes–Matuyama boundary is shown, as are units recording subchrons within the Matuyama. From this information, a proposed sequence of glaciations and interglaciations for Banks Island has been constructed (Table 2). The various lithostratigraphic units that document each event are grouped together. For example, marine or glaciomarine units unequivocally associated with a specific glaciation have been combined with the till of that glaciation. Most events were previously defined (Vincent 1983, 1984, 1989, 1990, 1992), but the paleomagnetic data now allow new events to be recognized.

Based on the magnetostratigaphic evidence presented for the four localities on Banks Island, we outline here the modifications to previous correlations of glacial and interglacial events given in Table 1. The discovery of normally magnetized paludal deposits (unit 2, section A) within an otherwise reversely magnetized preglacial Worth Point Formation has

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Fig. 12. Lithostratigraphic and magnetostratigraphic correlation of composite sections at the four studied localities.

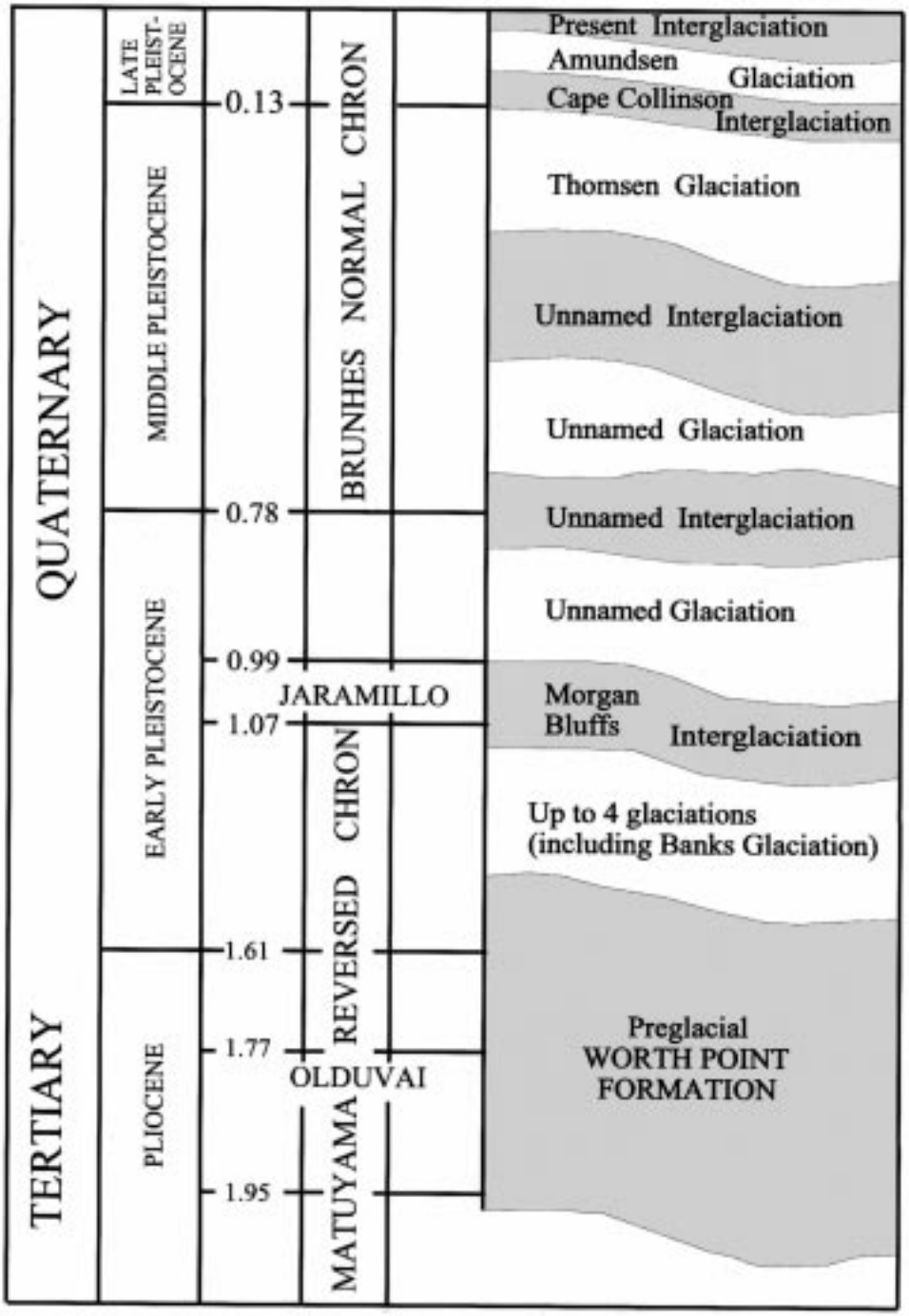


provided additional chronological control for the basal part of the Worth Point Bluffs. Based on arguments presented earlier, the paludal deposits have been assigned to the Olduvai subchron (Figs. 12, 13), and hence the Worth Point sediments are slightly older than previously estimated (Table 1). A second normal polarity subchron recorded in the upper part of interglacial sediments at Morgan Bluffs (unit 6, section C) postdates the preglacial Worth Point Formation and is assigned to the Jaramillo. As these sediments are overlain by reversely magnetized glacial sediments (unit 8, all sections), and interglacial sediments that span the Brunhes–Matuyama boundary (unit 5, sections A and C at Duck Hawk Bluffs), it was necessary to add an unnamed glaciation and interglaciation to the Early Pleistocene record on Banks Island (Fig. 13).

At both the Duck Hawk Bluffs and Nelson River Bluffs, normally magnetized interglacial sediments (unit 6, section H, and unit 8, section A, respectively) underlie deposits of the Thomsen Glaciation and late Middle Pleistocene Nelson River Formation (Table 1). These sediments cannot be associated with those of the Morgan Bluffs Interglaciation, which are reversely magnetized at the type section and predate the Nelson River Formation, and we therefore add an unnamed glaciation and interglaciation to the early Middle Pleistocene on Banks Island. Figure 13 shows all recognized glacial and

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Fig. 13. Proposed Late Tertiary and Quaternary preglacial, glacial, and interglacial events as recorded on Banks Island. Time scale in Ma.



interglacial events, set in a chronostratigraphic framework and places all events within the global reversal chronology.

Late Pliocene and Early Pleistocene (Matuyama Chron)

Following deposition of the fluvial coastal plain sediments of the Beaufort Formation, there was a preglacial interval, documented by sediments of the Worth Point Formation at Worth Point and Duck Hawk bluffs. It is assigned to the latest Late Pliocene or earliest Early Pleistocene based on the preglacial nature of the deposits, the generally reversed polarity, and the presence within it of normally magnetized paludal deposits most probably of Olduvai age.

The first arrival of continental ice, of mainland provenance, is recorded by the Bernard and other tills referred to the Banks Glaciation. These tills are present at all localities and are reversely magnetized. At Nelson River Bluffs, the tills may record as many as four distinct continental glaciations. It is not possible to determine which of the till sheets at Nelson River Bluffs corresponds to the single till sheet that occurs at Worth Point and Duck Hawk bluffs. It is interesting to note that on

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Banks Island, the first continental glaciation postdated the first glaciation in the Canadian Cordillera by at least a million years, the latter occurring sometime within the late Gauss (3.0–2.6 Ma; Barendregt et al. 1996; Duk-Rodkin and Barendregt 1997; Froese et al. 1997).

An interglaciation of Matuyama age is recognized at Morgan Bluffs, based on the presence of reversely magnetized fluvial sediments of the interglacial Morgan Bluffs Formation. Within this formation, there are normally magnetized fine-grained sediments occurring as fossil ice wedges, which probably belong to the Jaramillo subzone. These ice wedges were almost certainly developed on a tundra surface shortly after deposition of the Morgan Bluffs Formation.

If the assignments of the normally magnetized sediments of Worth Point Formation to the Olduvai subzone and those of the Morgan Bluffs Formation to the Jaramillo subzone are correct, then the Banks Glaciation(s) and the Morgan Bluffs Interglaciation date from the Early Pleistocene and their ages must fall between 1.77 and 1.07 Ma. A further consequence is that a substantial gap occurs in the Matuyama record at Worth Point Bluffs (within the upper part of unit 2, above the paludal member, Figs. 2 and 12).

At Morgan Bluffs, another glaciation of Matuyama age is recorded by a sequence of reversed glacial sediments postdating the Morgan Bluffs Interglaciation. This glaciation postdates beds we have assigned to the Jaramillo subchron and therefore probably occurred sometime between 0.99 and 0.78 Ma.

At Duck Hawk Bluffs, an interglaciation, which occurred at the end of the Matuyama Chron, is recorded in a sequence of predominantly reversely magnetized nonglacial sediments. At the top of this sequence, transitional directions and finally normal polarities are recorded. Hence, the Brunhes–Matuyama reversal (B–M, 0.78 Ma) occurs within sediments laid down during this interglaciation (between units 5 and 6, sections A and H, Figs. 3, 12).

These events are summarized as follows: (*i*) a period of nonglacial (presumably preglacial) sedimentation (<2.58 to <1.77 Ma); (*ii*) at least one (Banks Glaciation) but possibly as many as four continental glaciations (<1.77 to >1.07 Ma); (*iii*) an interglacial period referred to as the Morgan Bluffs Interglaciation (<1.77 to >0.99 Ma); (*iv*) an unnamed continental glaciation (<0.99 to >0.78 Ma); and (*v*) part of an unnamed interglaciation during which the B–M reversal occurred (about 0.78 Ma).

Middle Pleistocene (earlier Brunhes Chron)

Following the B–M reversal, a glaciation assigned to the early Middle Pleistocene is recorded by normally magnetized glacial sediments in the Nelson River Bluffs. Interglacial conditions, documented by terrestrial and perimarine sediments of normal polarity, are recorded at Duck Hawk and Nelson River bluffs and are assigned to the earlier part of the Middle Pleistocene. Deposits at these two localities may be correlative or they may represent two distinct interglaciations. Subsequently, Banks Island was again glaciated in the latter part of the Middle Pleistocene (Vincent 1992), as evidenced at all four localities by normally magnetized deposits of the Thomsen Glaciation.

These events are summarized as follows: (*i*) the last part of the interglaciation that records the B–M boundary (about 0.78 Ma); (*ii*) an unnamed glaciation (<0.78 Ma); (*iii*) one or more unnamed interglaciations (<0.78 to >0.13 Ma); and (*iv*) the Thomsen Glaciation (>0.13 Ma).

Late Pleistocene (later Brunhes Chron)

The Late Pleistocene record begins with interglacial sediments, which overlie Thomsen Glaciation deposits at the Duck Hawk and Nelson River localities, and which have been assigned to the Sangamon-age Cape Collinson Interglaciation (Vincent 1992). These are followed by sediments of the Amundsen Glaciation, and by Holocene sediments of the present interglaciation. All are normally magnetized. The Late Pleistocene is therefore represented by one glacial and two interglacial intervals, all younger than 0.13 Ma.

Mean direction and paleopole

The material as a whole is not well suited for accurate determination of geomagnetic field directions, nevertheless an attempt has been made to do this because of the paucity of Quaternary records from such high latitudes as Banks Island.

R* and N* magnetizations have been compiled and means for polarity zones at each locality calculated giving each specimen unit weight (Table 4). This procedure, although estimating mean directions quite well, almost certainly underestimates errors. In Table 4, an attempt is made to calculate means (grand averages) whose errors are more realistic. First, specimen data from lithologic units set out in Figs. 2, 4, and 5 are averaged, and the means of these averages taken (24 units at the three localities record N* and R* magnetizations). Second, averages are calculated for seven groups of lithologic units, termed major polarity units, and these are in turn averaged. The major polarity units, which are defined in Table 2, span approximately 100–200 ka and are considered sufficient to average out secular variation and provide mean directions which are not biased through our selective sampling of fine-grained units and the underrepresentation of coarser units. Statistically the two procedures give indistinguishable results. We regard the grouped averages of lithologic units as probably the best measures. Directional data from Duck Hawk Bluffs (see Table 4, data from Barendregt and Vincent 1990) are not included in the grand averages because the separation of R* and N* data could not be made from these older observations.

The means of the Brunhes and Matuyama fields fall short of being antiparallel by 11°, most of which can be accounted for by statistical errors. The difference is in a sense expected from the presence of unremoved, very small present field overprints. One result is quite clear: the average, irrespective of sign (in which the effect of residual overprints is removed), does not differ significantly from the geocentric axial dipole field (000, 80). Therefore, these sediments contain no inclination errors. The paleopole calculated from the average of the lithologic units ($n = 24$) is 86°N, 300°E (Fisher's estimate of precision $\kappa = 8$, and the half angle of the cone of confidence $\alpha_{95} = 11°$), which is not significantly different from the present geographic pole.

Summary of paleoclimatic history

Stratigraphic mapping and magnetostratigraphic studies have established a sequence of polarity zones and subzones and allowed the allocation of preglacial, glacial, and interglacial

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Table 4. Summary of directions.

Locality	Assigned magnetozone	n^a	D (°)	I (°)	k	α_{95} (°)
Nelson River Bluffs	Brunhes	144	47	78	16	3
Worth Point Bluffs	Brunhes	29	38	78	47	4
	Matuyama	7	180	–78	10	20
	Olduvai	4	101	85	98	9
	Mean	40	34	79	29	4
Morgan Bluffs	Brunhes	15	41	80	22	8
	Matuyama	78	146	–78	11	5
	Jaramillo	4	27	75	179	7
	Mean	97	337	79	12	4
Duck Hawk Bluffs	Brunhes	57	8	78	32	3
	Matuyama	80	176	–69	23	3
	Mean	137	359	73	24	3
Grand averages	Brunhes (lithologic units)	13	40	79	49	6
	Brunhes (major polarity units)	4	35	80	87	10
	Matuyama (lithologic units)	11	159	–79	14	13
	Matuyama (major polarity units)	3	159	–78	255	8
	All lithologic units	24	13	81	21	7
	All major units	7	9	80	85	7

Notes: Mean directions of N* and R* magnetizations are as defined in the text. D and I are the declination and inclination of the mean, respectively; k is Fisher's estimate of precision; and α_{95} is the half-angle of the cone of confidence ($P = 0.05$). Means for localities have been compiled irrespective of sign. As discussed in the text, mean directions for Duck Hawk Bluffs are not included in the grand averages. Mean directions for polarity subchrons are calculated from few samples and are therefore not included in grand average means. The paleopole calculated from the mean of all lithological units ($n = 24$) is 86°N, 300°E ($k = 8$, $\alpha_{95} = 11°$).

[a] For the four localities n is the number of individually oriented specimens to which unit weight is given; for the grand averages n is the number of lithologic or major polarity units.

events to the global reversal time scale and the classical subdivisions of the later Cenozoic (Table 2; Fig. 13). Table 1 gives the stratigraphic sequences recognized before the present study. Table 2 is an update of Table 1, based on extensive additional paleomagnetic sampling, and provides a revision of the stratigraphic and geological events framework established for sections on Banks Island. Figure 13 is a composite of all sections.

During the latest Late Pliocene and earliest Early Pleistocene (2.58 to about 1.77 Ma), nonglacial conditions prevailed. During this interval, Banks Island was underlain by continuous permafrost, and fluvial, aeolian, and colluvial processes modified the landscape. An open larch-dominated forest-tundra vegetation was present (Matthews et al. 1986; Vincent 1990). As climate deteriorated, the first of a series of continental glaciations affected Banks Island. This, the Banks Glaciation, may have comprised one or up to four glaciations during the interval 1.77–1.07 Ma. Then followed the Morgan Bluffs Interglaciation, which contains the Jaramillo subchron and dates from about 1.0 Ma. Before the Early Pleistocene ended, an unnamed glaciation reached Banks Island (0.99–0.78 Ma), and was followed by an interglaciation during which the Brunhes–Matuyama reversal occurred (0.78 Ma). During the Middle Pleistocene (0.78–0.13 Ma), an unnamed glaciation, an unnamed interglaciation, and the Thomsen Glaciation affected the landscape of Banks Island. Finally, during the Late Pleistocene (<0.13 Ma to present), the Cape Collinson Interglaciation (Sangamon), the Amundsen Glaciation, and the present-day interglaciation occurred.

Sediments on Banks Island provide a record of at least two and possibly as many as five glaciations and two interglaciations (in addition to a preglacial interval) during the Matuyama Chron. During the Brunhes Chron, three glaciations and three interglaciations are documented. This is one of the longest and most complete terrestrial records of climatic fluctuations in North America during the Quaternary.

The Brunhes–Matuyama reversal on Banks Island occurred during interglacial conditions. In deep-sea cores, the reversal occurs at oxygen-isotope stage 19 (De Menocal et al. 1990), a stage characterized as interglacial. This reversal has also been located between a reversely magnetized paleosol and a normally magnetized diamicton at three sections in the Kennedy Drift of Montana, and at two sections in the Mackenzie Mountains (Cioppa et al. 1995; Barendregt et al. 1991, 1996), suggesting that the field reversed during or after a period of soil formation (interglacial), but before the arrival of the next ice sheet (glacial). Therefore, it is very probable indeed that the reversal took place during interglacial times.

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